



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KOWRAPPA GOUNDER & SONS FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

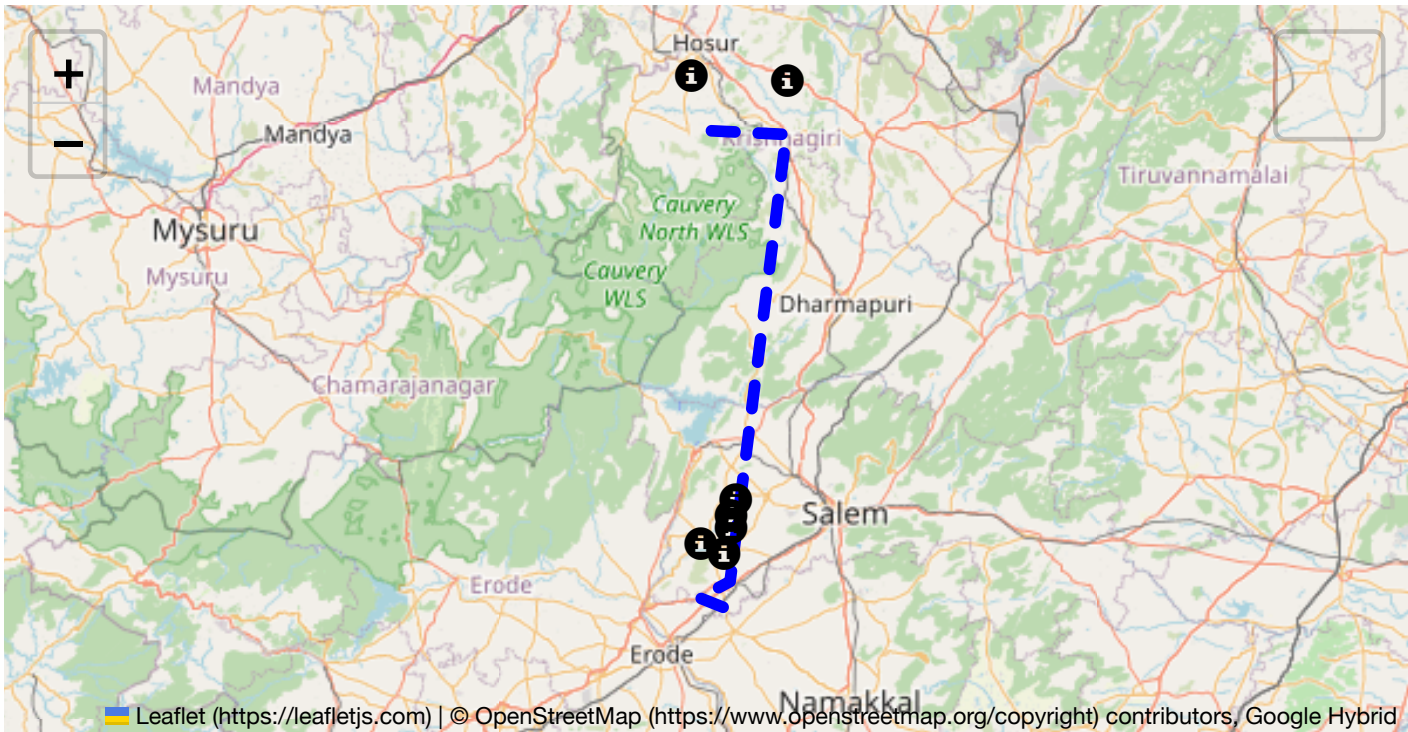
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 184.30 km
Estimated Duration: 3.8 hours

Adjusted Duration (Heavy Vehicle): 4.8 hours

Start: (11.4381, 77.8734)

End: (12.532177, 77.795348)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 184.3 kilometers, starting at Sangagiri and ending at Denkanikottai. Key waypoints include Padaiveedu, Ayveli, Olakkachinnanur, Erumaipatti, and Pillari Agraharam. The journey primarily traverses state highways and local roads, which may vary significantly in quality and congestion levels.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with significant rainfall during the monsoon season (October to December). The roads can become slippery and prone to flooding during heavy rains, particularly in low-lying areas. Travelers should prepare for reduced visibility due to fog in elevated areas, especially during the early morning hours.

3. Analysis of Traffic Patterns

Traffic can become particularly congested around major towns such as Sangagiri and Denkanikottai during peak hours (8 AM to 10 AM and 5 PM to 7 PM). Additionally, areas near educational institutions and marketplaces may witness higher pedestrian activity and slower traffic movement.

4. Assessment of Road Quality and Infrastructure

The road quality varies across the route, with well-maintained sections along major state highways but potential deterioration in rural and interior roads. Watch out for potholes and narrow roads, which could be challenging for large trucks, especially in areas that see frequent heavy vehicle usage.

5. Suggestions for Alternative Routes for Emergencies

In case of closures or exigencies, consider using the NH44, connecting major districts before diverting to smaller state highways. This alternative might slightly increase the travel distance but offers better-maintained routes, more emergency services, and amenities.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials requires adherence to local and national regulations, including adherence to time restrictions for movement, ensuring proper signage on the transport vehicle, and having necessary permits and documentation.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While specific incident records are limited, the region has reported typical issues such as vehicle breakdowns and minor accidents due to road and weather conditions. Hazardous material incidents may not be frequent, but caution is advised around populated areas.

8. Environmental Considerations and Sensitive Areas

The route traverses agricultural lands and near wildlife sanctuaries. Therefore, drivers should be cautious to avoid spillages and ensure that emissions are within regulated limits to minimize environmental impact.

9. Analysis of Communication Coverage

Communication coverage is generally good along major highways and urban areas but can be patchy in rural zones. It's advisable to have a secondary form of communication or a pre-planned communication schedule in place.

10. Estimated Emergency Response Times

Emergency response times vary, with urban areas potentially seeing responders within 20-30 minutes. In rural and remote stretches, it may take over an hour depending on the nearest emergency service location.

12. Overall Summary of Risk Assessment

This route presents a moderate risk level primarily due to varying road quality, potential weather impacts, and the presence of populated and rural areas. Careful planning with contingencies for weather, traffic, and communication issues is crucial. Ensuring vehicles are in optimal condition and drivers are well-briefed on local conditions will mitigate many of the risks identified.

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.60628, 77.86014	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	Medium	11.43995, 77.87431	30 KM/Hr
3	Turn	High	11.44107, 77.87528	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	Medium	11.45781, 77.85203	30 KM/Hr
7	Turn	High	11.46307, 77.84941	15 KM/Hr
8	Turn	Medium	11.45546, 77.81800	30 KM/Hr
9	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
10	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
11	Turn	High	11.49469, 77.88488	15 KM/Hr
12	Turn	Medium	11.57662, 77.90612	30 KM/Hr
13	Turn	High	11.59235, 77.90675	15 KM/Hr
14	Turn	Medium	11.59449, 77.91304	30 KM/Hr
15	Turn	Medium	11.63843, 77.94821	30 KM/Hr
16	Turn	High	11.64192, 77.95195	15 KM/Hr
17	Turn	Medium	11.64747, 77.95129	30 KM/Hr
18	Turn	Medium	11.67875, 77.96855	30 KM/Hr
19	Turn	High	11.70276, 77.97907	15 KM/Hr
20	Turn	Medium	11.71380, 77.99532	30 KM/Hr
21	Turn	Medium	11.71731, 78.00006	30 KM/Hr
22	Turn	Medium	11.71947, 78.01172	30 KM/Hr
23	Turn	Medium	11.72760, 78.02285	30 KM/Hr
24	Turn	High	11.73785, 78.03496	15 KM/Hr
25	Turn	High	11.73653, 78.03886	15 KM/Hr
26	Turn	High	11.74252, 78.04705	15 KM/Hr
27	Turn	Medium	11.77998, 78.05786	30 KM/Hr
28	Turn	Medium	11.86915, 78.08382	30 KM/Hr
29	Turn	Medium	11.87998, 78.08170	30 KM/Hr
30	Turn	Medium	11.91266, 78.05852	30 KM/Hr
31	Turn	Medium	11.92653, 78.06517	30 KM/Hr
32	Turn	Medium	11.93196, 78.06473	30 KM/Hr
33	Turn	Medium	11.93473, 78.06268	30 KM/Hr
34	Turn	Medium	11.93746, 78.05331	30 KM/Hr
35	Turn	Medium	11.95531, 78.04698	30 KM/Hr
36	Turn	High	11.95854, 78.04990	15 KM/Hr
37	Turn	High	11.96015, 78.04945	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	Medium	11.96266, 78.05115	30 KM/Hr
39	Turn	High	11.96433, 78.05113	15 KM/Hr
40	Turn	High	11.96542, 78.04938	15 KM/Hr
41	Turn	Medium	12.12036, 78.13284	30 KM/Hr
42	Blind Spot	Blind Spot	12.51393, 78.03186	10 KM/Hr
43	Turn	Medium	12.51379, 78.02933	30 KM/Hr
44	Turn	Medium	12.51624, 78.02577	30 KM/Hr
45	Turn	Medium	12.52269, 78.02290	30 KM/Hr
46	Turn	Medium	12.52651, 78.01023	30 KM/Hr
47	Turn	High	12.55108, 77.97248	15 KM/Hr
48	Turn	Medium	12.55286, 77.96596	30 KM/Hr
49	Turn	Medium	12.56524, 77.95283	30 KM/Hr
50	Turn	Medium	12.56246, 77.92215	30 KM/Hr
51	Turn	Medium	12.57304, 77.89875	30 KM/Hr
52	Turn	Medium	12.58211, 77.89398	30 KM/Hr
53	Turn	Medium	12.58409, 77.88837	30 KM/Hr
54	Turn	High	12.58358, 77.88760	15 KM/Hr
55	Turn	Medium	12.59952, 77.86311	30 KM/Hr
56	Turn	High	12.60497, 77.85468	15 KM/Hr
57	Turn	Medium	12.59990, 77.85037	30 KM/Hr
58	Turn	Medium	12.59351, 77.84812	30 KM/Hr
59	Turn	Medium	12.59167, 77.84620	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
4	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
9	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
11	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
13	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
22	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
27	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
29	clinic	Kowthalam HSC	12.5676496, 77.8312573	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
23	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
30	school	St. Jopseph School	12.5339473, 77.795134	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.60628, 77.86014



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44107, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57662, 77.90612



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59235, 77.90675



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59449, 77.91304



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63843, 77.94821



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64192, 77.95195



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64747, 77.95129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67875, 77.96855



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70276, 77.97907



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71380, 77.99532



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71731, 78.00006



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71947, 78.01172



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72760, 78.02285



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92653, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93196, 78.06473



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93473, 78.06268



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



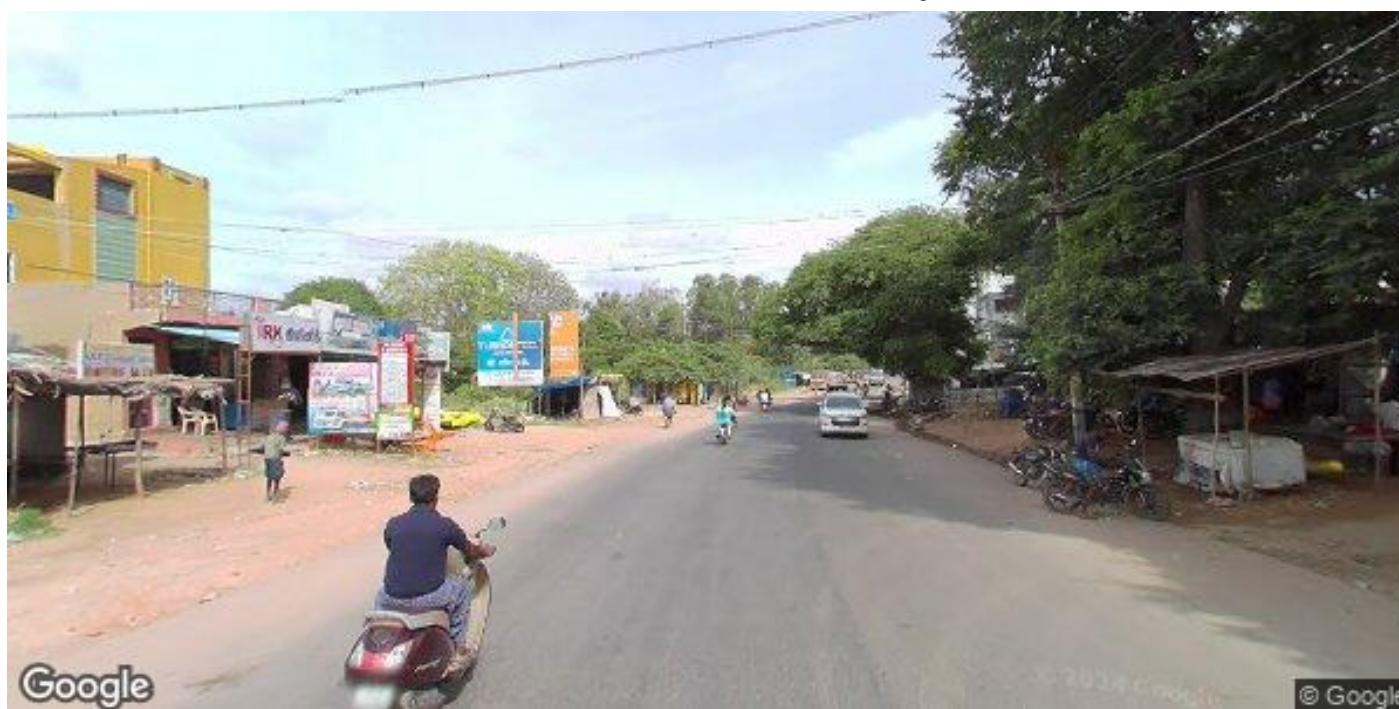
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12036, 78.13284



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51393, 78.03186



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51379, 78.02933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51624, 78.02577



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52269, 78.02290



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52651, 78.01023



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.55108, 77.97248



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55286, 77.96596



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56524, 77.95283



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56246, 77.92215



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.57304, 77.89875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58211, 77.89398



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58409, 77.88837



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.58358, 77.88760



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59952, 77.86311



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.60497, 77.85468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59990, 77.85037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59351, 77.84812



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59167, 77.84620

